

KLH (DEEMED TO BE UNIVERSITY)

PROJECT REVIEW – 3 | TEAM 18 | SEC 10

Food Donation & NGO Connect Platform

A location-aware network connecting surplus food to NGOs



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Under the guidance of D. Divya Priya • Database Management Systems • A.Y. 2026–2027

Project Overview

THE PROBLEM

- ✗ Coordination runs on calls and WhatsApp groups
- ✗ No shared record of what is available or needed
- ✗ Matching donors to NGOs is slow and manual
- ✗ No tracking from pickup to delivery

OUR SOLUTION

- ✓ One platform for donors, NGOs and volunteers
- ✓ SQL matching by distance, quantity and expiry
- ✓ Triggers alert NGOs the moment food is listed
- ✓ Pickup-to-delivery tracking with an audit trail

TECH STACK

React · HTML5 · CSS3

Spring Boot

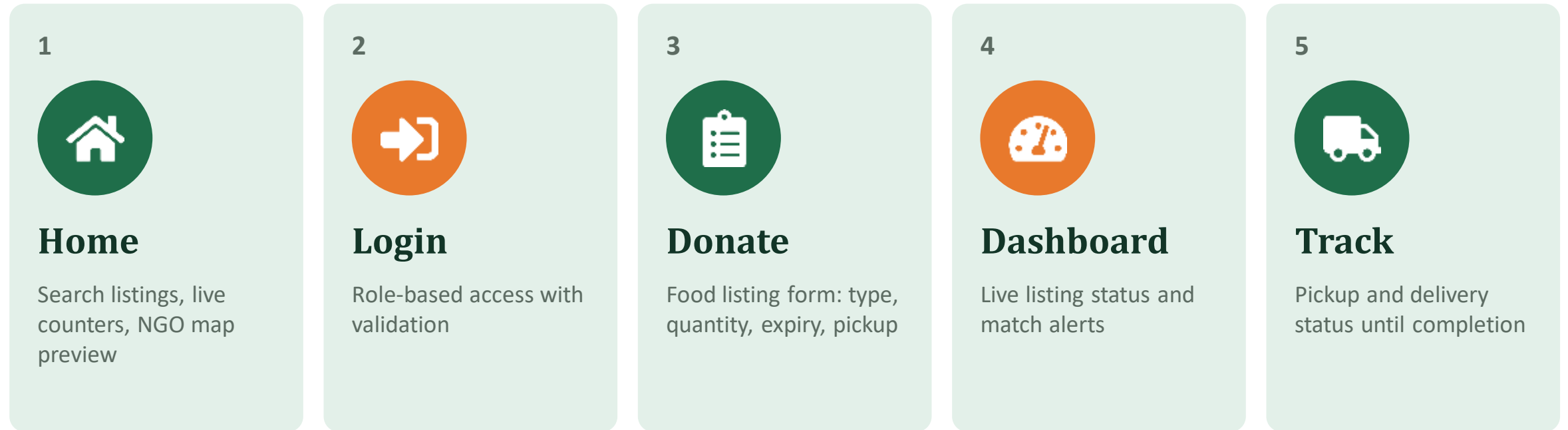
REST APIs

MySQL + JPA

Git & GitHub

Front-End: User Journey

How a user moves through the React-based interface



COMMON TO EVERY PAGE

Reusable header/footer • Responsive layout • Inline form validation • fetch() calls to REST APIs

Frontend Modules & Features



Post Donation

Food type, quantity, expiry, pickup address & photo

Used by: Donor



Browse & Filters

Find listings by food type, distance or expiry

Used by: NGO, Volunteer



My Dashboard

Track listing status and view match alerts

Used by: Donor, NGO



Pickup Tracker

Assign volunteers; update pickup & delivery status

Used by: NGO, Volunteer



NGO Requirements

Register capacity, needs and service area

Used by: NGO

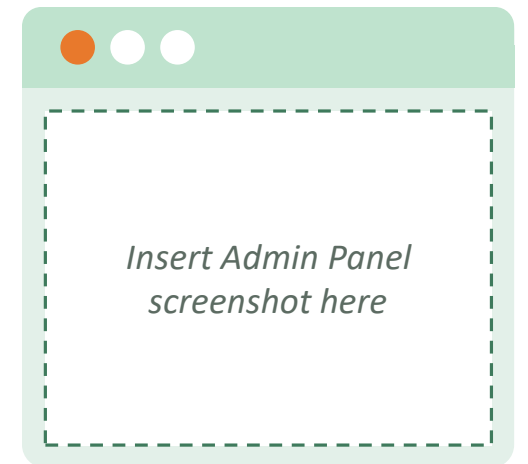
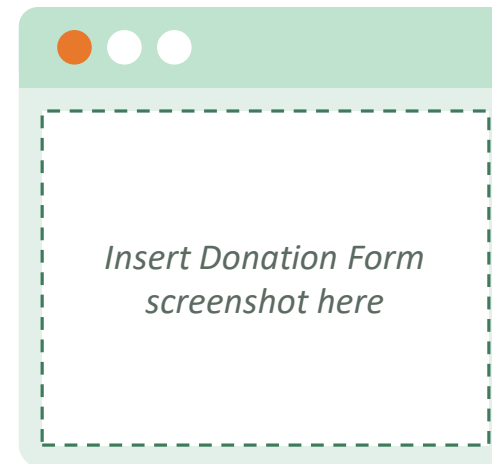
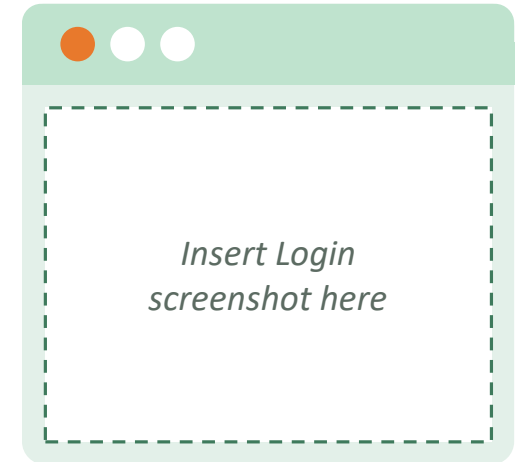
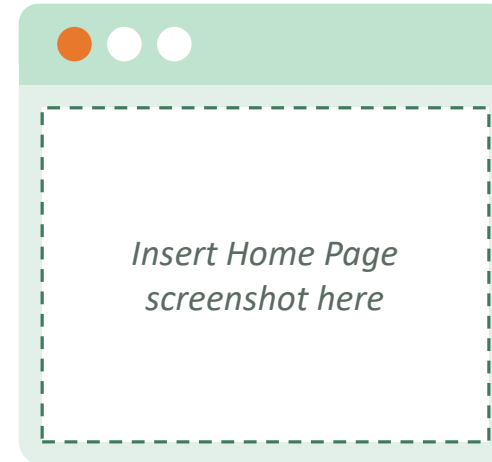
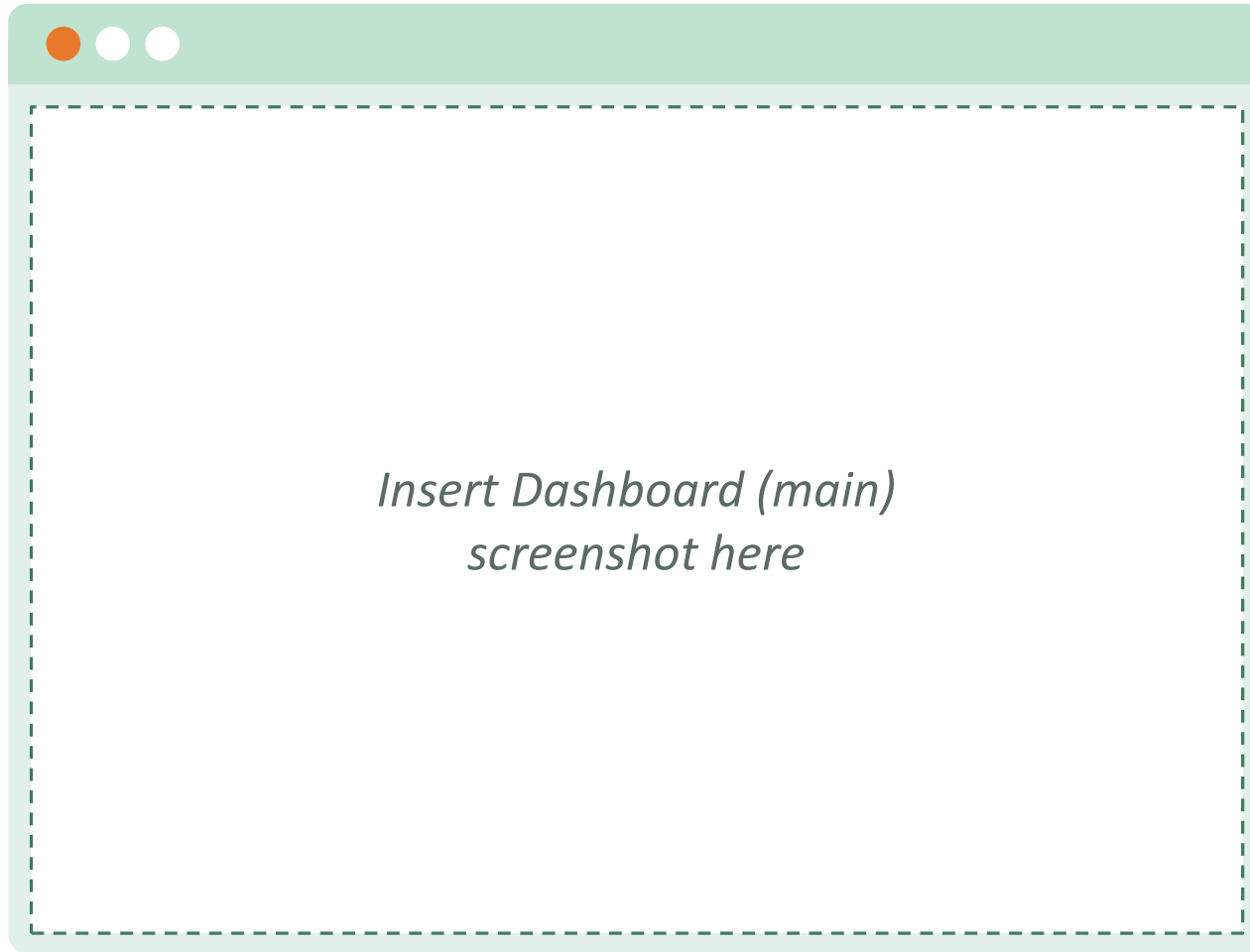


Admin Panel

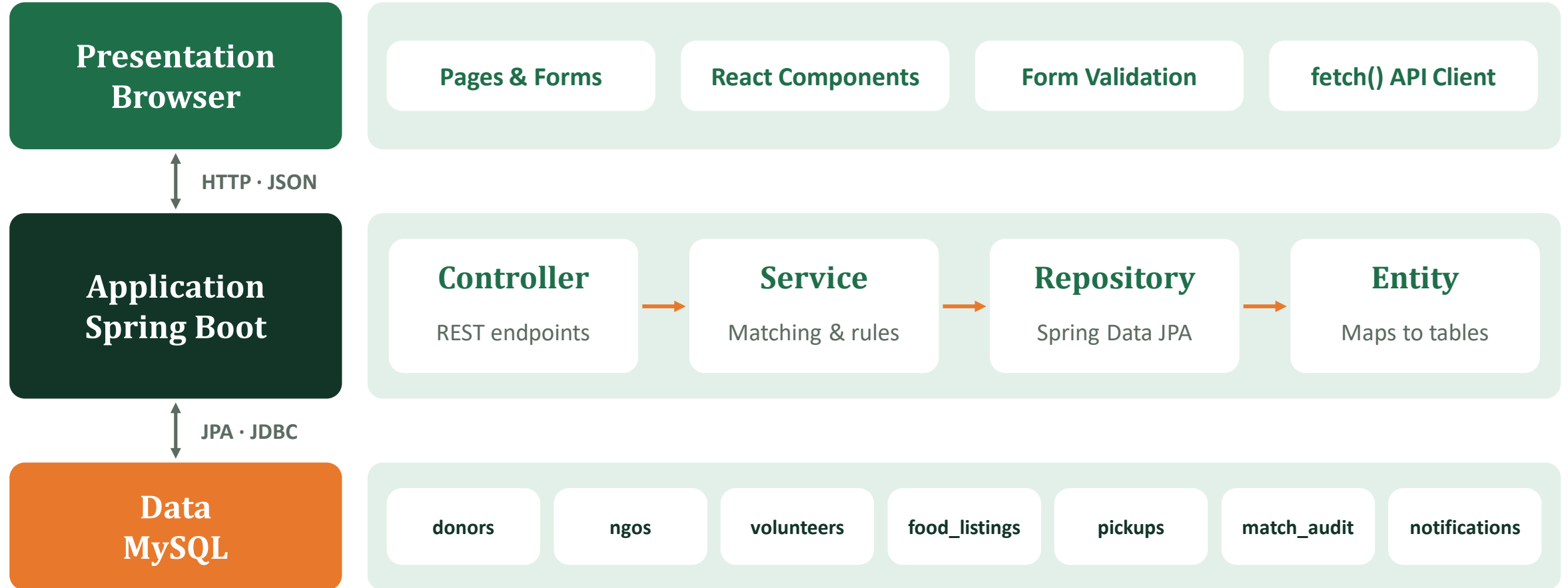
Verify users, resolve disputes, remove duplicates

Used by: Admin

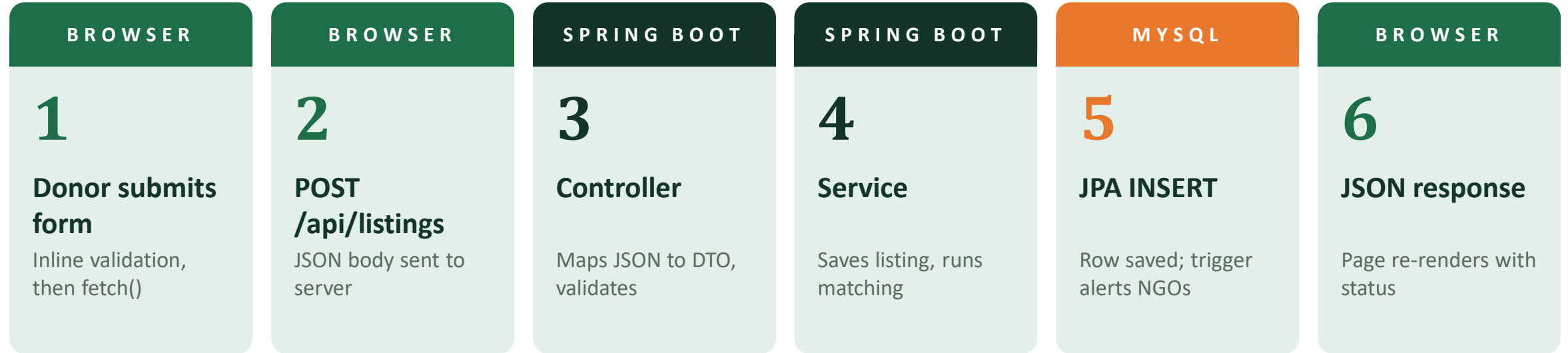
Frontend Screenshots



Architecture: Frontend → Backend → Database



Request Data Flow: Posting a Donation



REQUEST · POST /api/listings

```
{
  "donorId": 7,
  "foodType": "Cooked meals",
  "quantityKg": 25,
  "expiry": "2026-10-03T21:00"
}
```

RESPONSE · 201 CREATED

```
{
  "listingId": 214,
  "status": "AVAILABLE",
  "matchedNgos": 3
}
```

Database Design

7

relational tables

PK / FK keys link every table

Normalized up to 3NF

MySQL via Spring Data JPA

Why a relational design?



Donors, NGOs, listings and pickups are naturally linked records



Constraints keep every listing, match and pickup valid

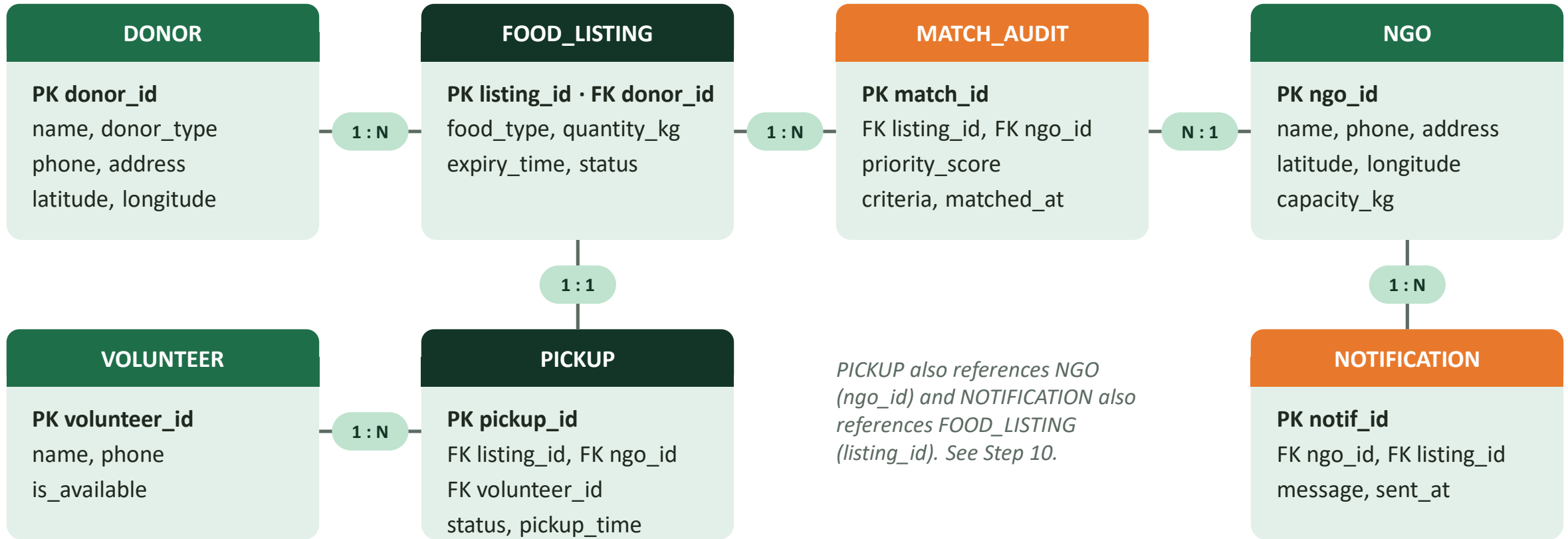


Each fact is stored once and referenced, not copied



Joins power proximity matching; triggers and indexes keep it fast

ER Diagram / Database Schema



Tables and Their Purpose



CORE DATA

donors

Restaurants, households & event organizers

ngos

Verified NGOs with capacity and location

volunteers

Pickup volunteers and availability



PROCESS

food_listings

Surplus food details, expiry and status

pickups

Volunteer assignment, pickup & delivery tracking



SUPPORT

match_audit

Every match with priority score & criteria

notifications

Trigger-generated alerts to NGOs

Keys and Relationships

Primary key

Uniquely identifies a row

Foreign key

Points to another table's primary key

Referential integrity

No listing, match or pickup can point to a missing row

FOREIGN KEY → PRIMARY KEY

food_listings.donor_id



donors.donor_id

match_audit.listing_id



food_listings.listing_id

match_audit.ngo_id



ngos.ngo_id

pickups.listing_id



food_listings.listing_id

pickups.ngo_id



ngos.ngo_id

pickups.volunteer_id



volunteers.volunteer_id

notifications.ngo_id



ngos.ngo_id

notifications.listing_id



food_listings.listing_id

Database Operations (CRUD)



CREATE

SQL INSERT

REST POST

Post a donation or register an NGO



READ

SQL SELECT

REST GET

View listings, matches, dashboard



UPDATE

SQL UPDATE

REST PUT

Change status → Matched / Picked Up / Delivered



DELETE

SQL DELETE

REST DELETE

Admin removes duplicate or expired listings

Sample SQL Queries

1 Available listings (not expired)

```
SELECT * FROM food_listings
WHERE status = 'AVAILABLE'
      AND expiry_time > NOW();
```

2 NGOs near a listing with enough capacity

```
SELECT n.name, l.food_type, l.quantity_kg
FROM food_listings l
JOIN donors d ON d.donor_id = l.donor_id
JOIN ngos n ON n.capacity_kg >= l.quantity_kg
WHERE l.listing_id = 214
      AND ABS(n.latitude - d.latitude) < 0.05
      AND ABS(n.longitude - d.longitude) < 0.05;
```

3 Best matches first

```
SELECT n.name, l.food_type, m.priority_score
FROM match_audit m
JOIN food_listings l ON l.listing_id = m.listing_id
JOIN ngos n ON n.ngo_id = m.ngo_id
ORDER BY m.priority_score DESC;
```

Triggers, Procedures & Indexing



Trigger

Fires after every new listing and starts matching automatically.

```
CREATE TRIGGER trg_notify_ngos
AFTER INSERT ON food_listings
FOR EACH ROW
  CALL sp_match_listing(
    NEW.listing_id);
```



Stored Procedure

Reusable matching rules kept inside the database.

```
PROCEDURE sp_match_listing(id)
-- 1. filter NGOs by distance
-- 2. check capacity, expiry
-- 3. score, INSERT match_audit
-- 4. INSERT notifications
```



B-Tree Index

Fast lookup of nearby, unexpired listings.

```
CREATE INDEX idx_listing_expiry
ON food_listings(expiry_time);
CREATE INDEX idx_ngo_location
ON ngos(latitude, longitude);
```

Conclusion and Next Steps

WHAT WE ACHIEVED



One platform connecting donors, NGOs and volunteers



Linked relational database in 3NF with keys, constraints and triggers



Frontend connected to the backend through REST APIs

NEXT STEPS

1

Improve matching accuracy and scoring

2

Integrate maps for live proximity

3

Add email / SMS alerts to NGOs

4

More validation & testing

5

Ratings and a verified trust network



Thank You

Food Donation & NGO Connect Platform

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